AI Acceptance Across Service Platforms

Text Mining Reddit and Trustpilot: Methods in Practice

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1 The idea and research question

Online services increasingly place AI agents between the user and the thing they came for. Chatbots answer support tickets, algorithms screen and price, automated systems take decisions. We study how people accept or reject that AI, and how acceptance changes with the role AI plays. We compare three service contexts:

- **AI-native services**, where the product itself is an AI agent (ChatGPT, Claude, Replika, Gemini, Perplexity);
- **Peer-to-peer rentals**, where AI is an automation layer on a human marketplace (Turo, Airbnb, Fat Llama);
- **Customer service**, where AI runs the support layer (chatbots replacing agents).

The middle context is the assignment's core: a peer-to-peer sharing economy for high-value professional gear such as cameras and drones (Fat Llama), photography lenses (Lensrentals) and film and event kit (KitSplit), alongside vehicles (Turo, Getaround) and RVs (Outdoorsy). The AI-native and customer-service contexts act as comparison points that show how acceptance shifts as AI moves from being the product, to a marketplace add-on, to the support desk.

> **Research question.** How do users accept and experience AI agents across different service platforms, and what drives or erodes trust as AI moves from being the product, to a marketplace add-on, to the support desk?

Reddit is our discussion baseline, where users describe AI experiences in their own words. Trustpilot is our verification layer, where every review carries a 1 to 5 star rating that acts as a ground-truth acceptance label. Where the open discussion and the labelled reviews agree, we gain confidence. Where they diverge, we learn something.

2 The data

We collected two text sources and used no X or Twitter content.

Reddit (baseline). 4,269 comments from 12 subreddits (November 2023 to May 2026), collected from the Arctic Shift research archive (`01_fetch_reddit_arcticshift.R`) because Reddit's 2026 anti-bot wall blocks live scraping. AI-native subreddits are pulled in full; rental and support subreddits are pulled by AI and automation keywords. Every comment carries a `platform_category`: ai_service 2027, rental 1715, customer_service 527.

Trustpilot (verification). 3,912 reviews from seven peer-to-peer rental platforms (Turo, Fat Llama, RVshare, Outdoorsy, Lensrentals, Getaround, KitSplit), collected with an Apify actor and merged by

02_merge_trustpilot.R. Each review has a 1 to 5 star rating, and 06_ai_acceptance.R flags the ones that describe an AI or automation experience.

3 Method

All analysis runs in one script, analysis.R, after data collection. Reddit is mined in depth (steps 1 to 7) and Trustpilot enters as the verification layer (steps 8 to 10). The two classes compared throughout are positive versus negative acceptance: on Reddit inferred from a sentiment lexicon, on Trustpilot read directly from the star rating. We picked methods we could apply meaningfully to this question, and the pipeline covers every technique the brief asks for.

Table 1: The analysis pipeline (one script, analysis.R).

Step	Method	Why
1 Preprocess + classes	Lemmatise + AFINN lexicon split	Normalise text; build the two classes
2 Word frequency	Top-10 words per class	What each class is about
3 Word clouds	Commonality + comparison, 1/2-gram	Shared vs distinctive words
4 Sentiment	NRC eight emotions	Emotional texture beyond polarity
5 Co-occurrence network	Bigram network (igraph, ggraph)	Which words travel together
6 Topic models (LDA)	LDA, 4 topics per class, 1/2-gram	Latent themes per class
7 Word embeddings	GloVe word vectors	Word meaning and neighbours
8 Validation	Inferred class vs star rating	Test labels against real stars
9 Themes + role gradient	Shared theme tagging, both sources	Frustrations and the role of AI

For the positive versus negative split we use the **AFINN lexicon** (Nielsen, 2011), which scores words from minus 5 to plus 5; summing the scores in a comment and taking the sign gives its class. The text mining follows the tidy-text approach of Silge and Robinson (2017). AFINN downloads once through the textdata package and is then cached.

4 Findings

4.1 What each class talks about (word frequency and clouds)

After normalising the text (lowercasing, removing punctuation, numbers and stop words, then lemmatising to base words), we count the most frequent words in each class. Frequency is the fastest read on what each class is about. The negative class centres on *host, review, turo, guest, bot*; the positive class on *airbnb, host, bot, guest, contact, automate* (Figure 1). Both classes are dominated by rental-hosting vocabulary, since the rental subreddits are the largest. The word *automate* appears in both classes: hosts praise the automation that saves them time, and it also surfaces when automation frustrates. The comparison cloud (Figure 2) shows the words that most separate the two classes.

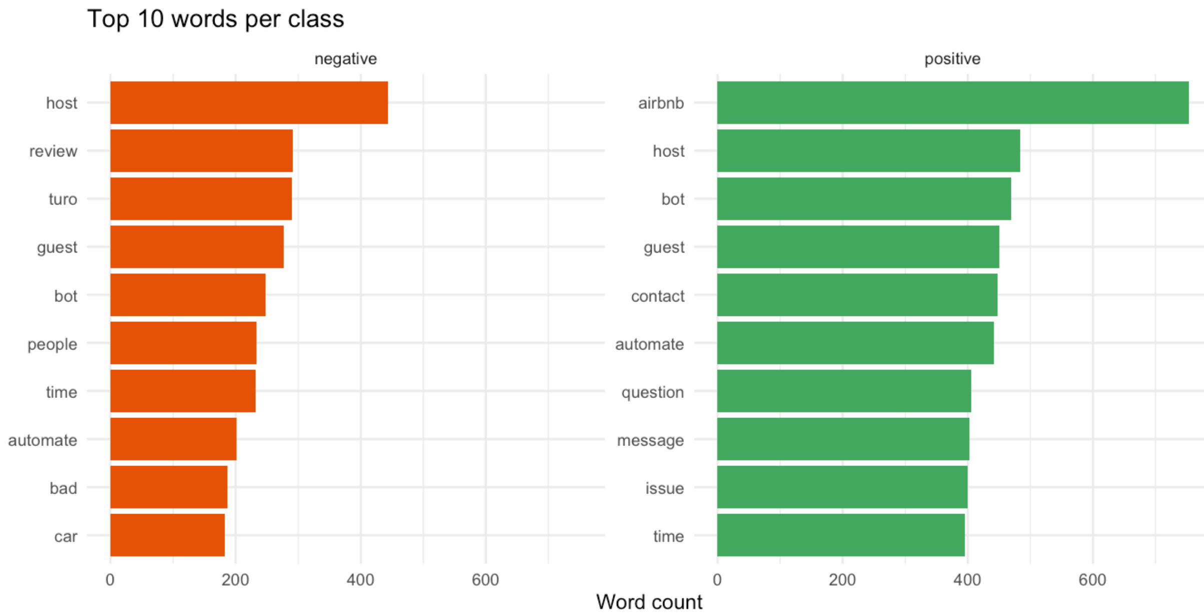

Figure 1: Top 10 words per class.

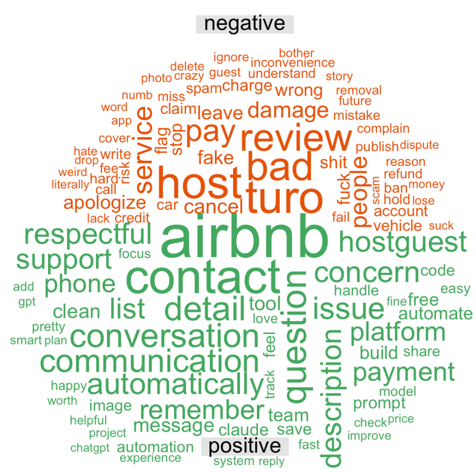

Figure 2: Comparison cloud: words coloured by the class they belong to.

4.2 How the classes feel (sentiment)

The two classes are split by polarity, so to add information we score the **NRC emotion lexicon** (Mohammad and Turney, 2013), which tags words with eight emotions plus an overall polarity. The positive class scores higher on trust, anticipation and joy; the negative class higher on sadness, fear, anger and disgust (Figure 3). The gap is widest on **trust**, which is the currency of a sharing economy.

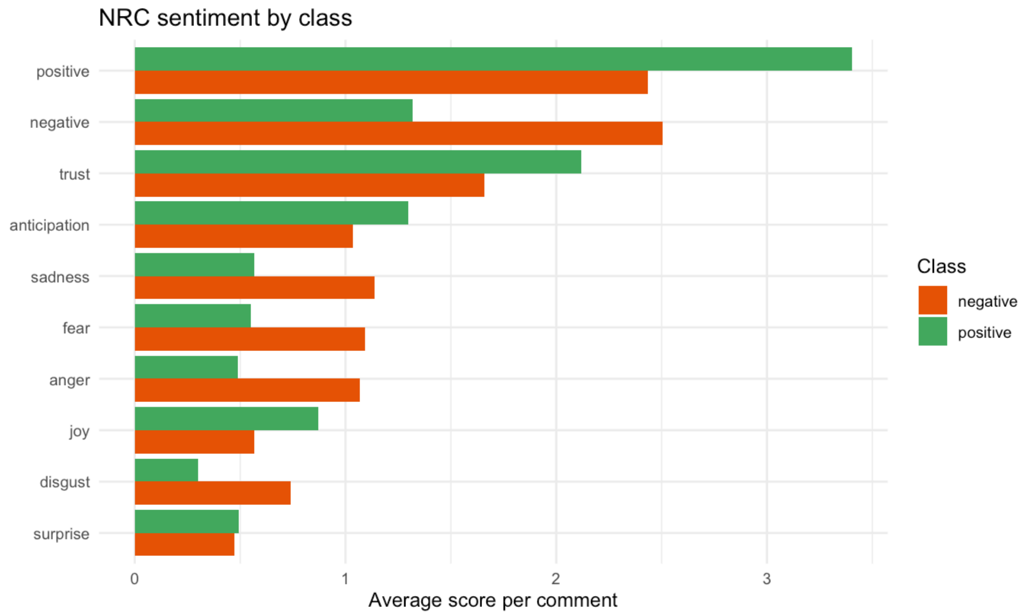

Figure 3: Average emotion words per comment, by class (NRC lexicon).

4.3 How words connect (co-occurrence network)

Frequency lists lose structure, so we count adjacent word pairs (bigrams) and draw the strongest as a network with igraph and ggraph. A network shows which words travel together. The network falls into clear clusters (Figure 4): a customer-support core (*customer, service, support, contact, question, issue*), a bot cluster (*bot, phone, chat, automatically, wait*), a host-automation cluster (*automate, message, guest, host, send*), and a review-moderation cluster (*review, post, manual, flag, spam, publish*) that reflects Airbnb's automated review handling.

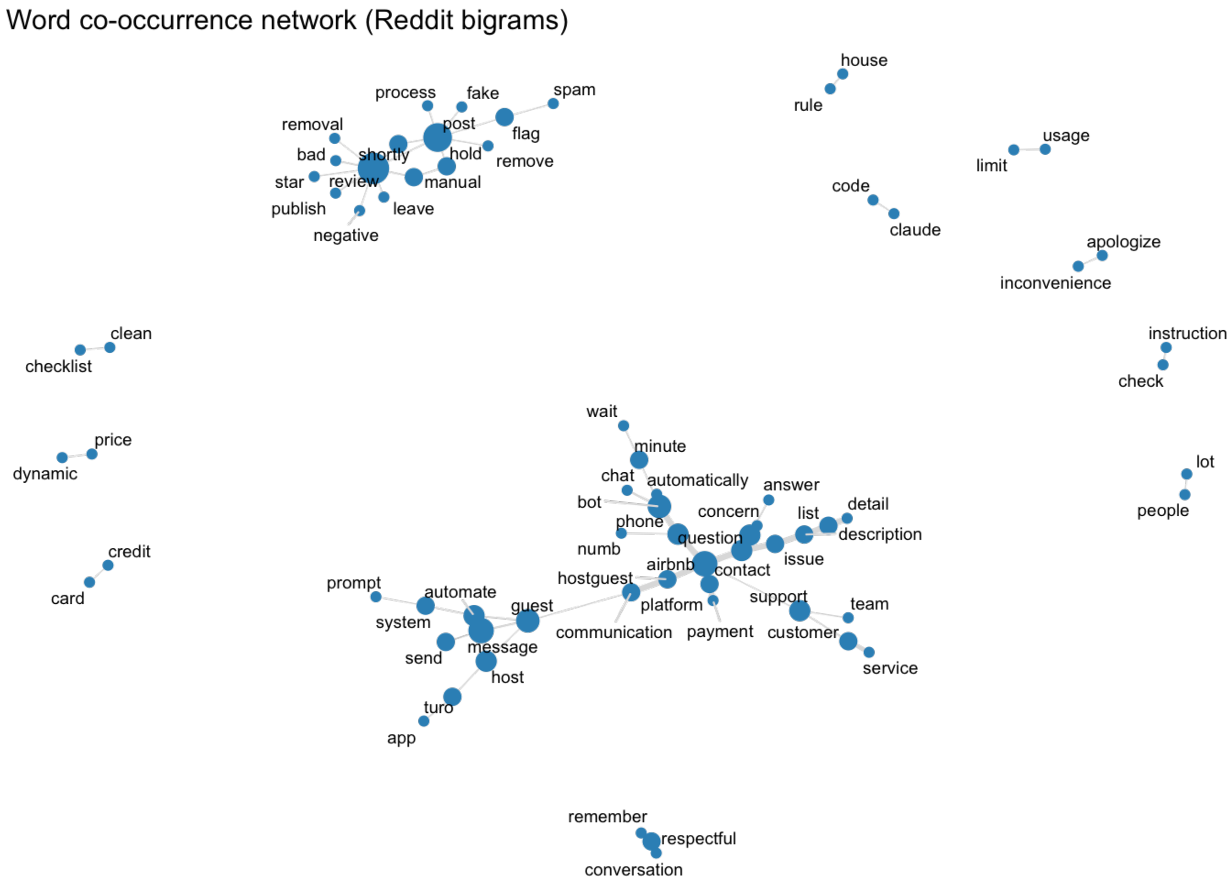

Figure 4: Word co-occurrence network of the strongest Reddit bigrams.

4.4 The latent topics (LDA)

Latent Dirichlet Allocation (Blei, Ng and Jordan, 2003) groups co-occurring words into topics. We fit four topics per class on unigrams and bigrams, which summarises thousands of comments into a few readable subjects. For the negative class (Table 2 and Figure 5), Topic 1 is content and complaints (*people, pay, model, post, write*), Topic 2 is bots and human support (*bot, customer, human, question, call, agent, chat*), Topic 3 is Airbnb hosting and reviews (*host, review, guest, airbnb, automate*), and Topic 4 is Turo car rentals (*turo, car, app, service, cancel*). Bots, support and reviews recur across topics.

Table 2: LDA topics for the negative class (top terms per topic).

Topic.1	Topic.2	Topic.3	Topic.4
people	bot	host	turo
pay	customer	review	car
model	human	guest	host
post	post	airbnb	time
stop	time	message	app
write	question	automate	service
bad	people	stay	cancel
reason	call	book	vehicle
issue	agent	photo	trip

Topic.1	Topic.2	Topic.3	Topic.4
answer	chat	bad	send

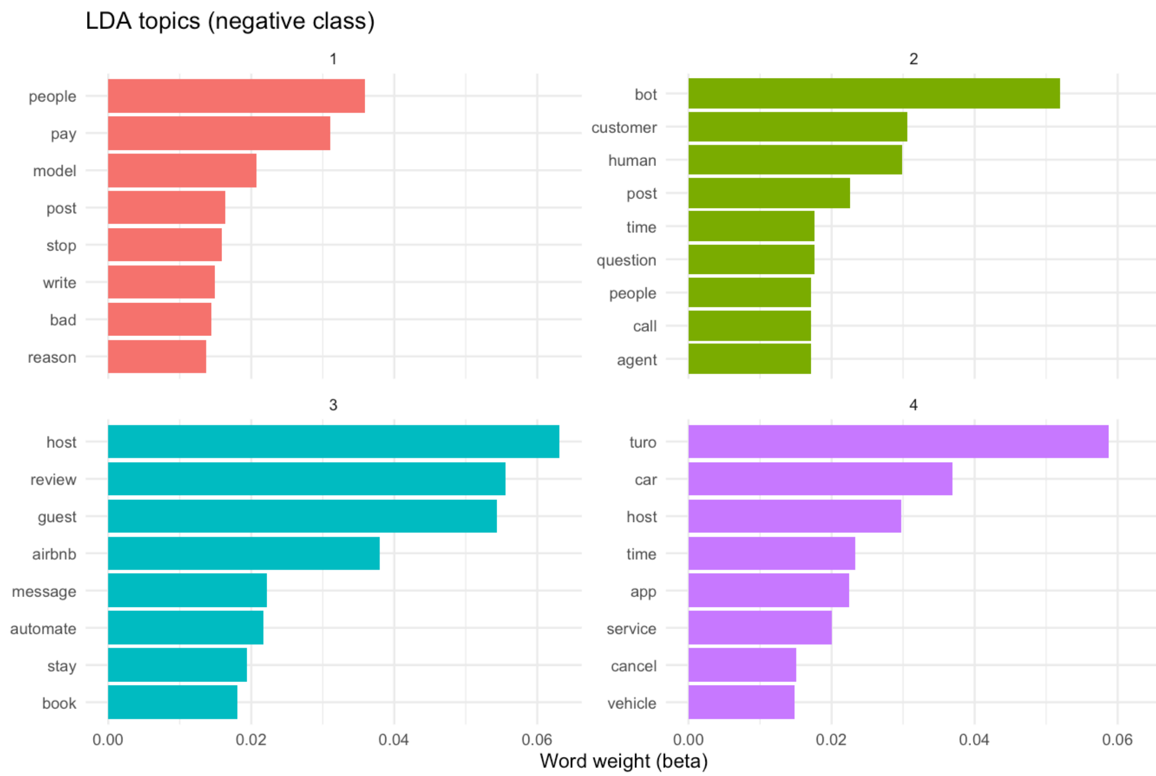

Figure 5: LDA topics for the negative class: top terms weighted by beta, the probability of the word within the topic.

4.5 The meaning behind words (embeddings)

Lexicon and topic methods ignore meaning. **GloVe** word embeddings (Pennington, Socher and Manning, 2014) place words in a vector space learned from the corpus, so we can read the nearest neighbours of key terms. The vectors independently rediscover the human-versus-automation tension at the heart of the project:

- **trust**: fix, stop, able, safety, dismiss, appointment, work, error
- **bot**: automatically, question, contact, phone, please, chat, issue, concern
- **human**: agent, without, support, actually, actual, need, handle, interaction
- **host**: guest, us, know, turo, also, stay, many, even
- **automate**: message, guest, system, set, send, response, place, us

The neighbours of *human* (agent, support, interaction, actual, need) point to users wanting a real person, while *bot* sits among contact channels (automatically, phone, contact, chat, issue). Projecting the vectors of the key terms down to two dimensions (Figure 6) makes the same geometry visible at a glance: *host, guest, review* and *automate* sit together as the hosting toolbox, *human, customer, support* and *trust* form their own corner, and *bot* sits apart from both.

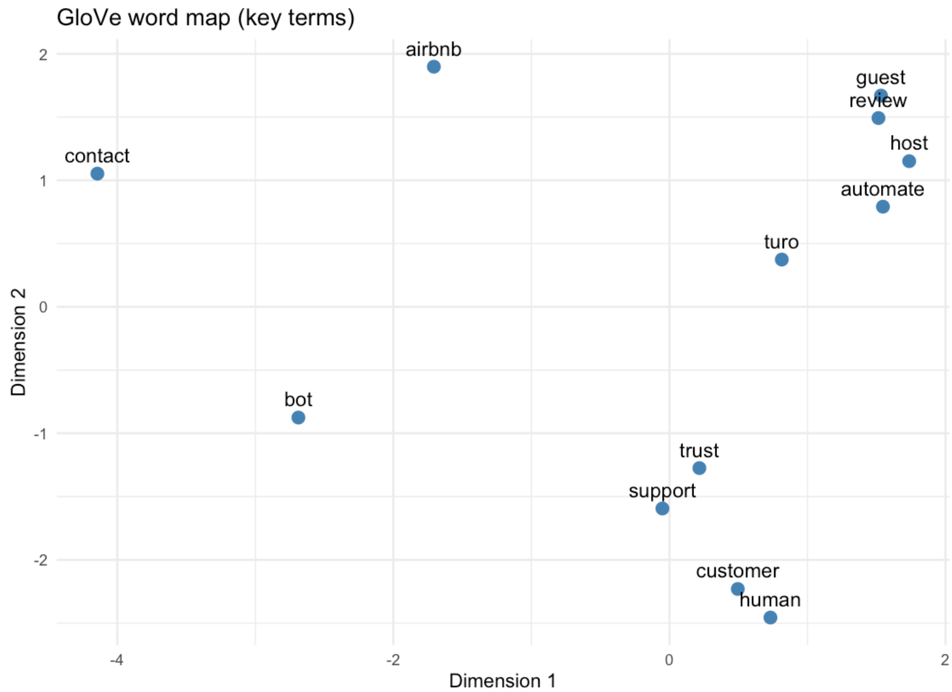

Figure 6: GloVe word map: key terms projected to two dimensions with PCA. Distance reflects how differently the corpus uses the words.

4.6 Does the inferred class hold up? (validation)

Reddit has no ground-truth label, so the positive or negative class is *inferred* from sentiment. Trustpilot has both text and a star rating, so it is where we can test that inference: we apply the same lexicon class to Trustpilot and compare it to the star-based class. Agreement is **92.8%** (Cohen's $\kappa = 0.74$, n = 3,544), which is substantial by the usual benchmark, and the lexicon recovers about 71% of genuinely negative reviews (Figure 7). This is the bridge that lets us read the two corpora together with confidence.

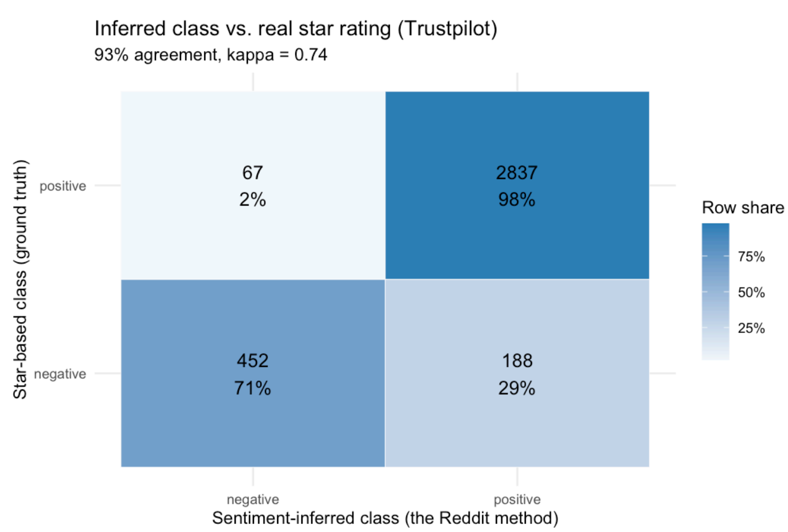

Figure 7: Inferred class vs. the real star rating on Trustpilot.

4.7 The big pattern (the role of AI)

With the method validated, we tag both corpora with the same six themes and measure acceptance by AI's role. On Reddit the three contexts are close, with AI-native services slightly the most positive:

Table 3: Reddit: positive-class share by the role of AI.

Context	Comments	Positive share
AI-native (the product)	1349	64%
Rental (a marketplace add-on)	1425	62%
Customer service (support desk)	425	61%

Because the rental and customer-service subreddits were collected by keyword search, those slices also contain general platform talk (almost all of it from the deliberately broad *app* query on r/Turo). As a robustness check we recompute the gradient using only comments that explicitly mention AI, automation, bots or a named AI product. The ordering is unchanged and the spread slightly wider: 65.9% for AI-native, 63.5% for rental and 61.5% for customer service (n = 2287) positive.

The decisive role signal is on Trustpilot: rental reviews that mention AI or automation average **2.32 stars** against **4.21** for the rest, a **1.89-star** gap (Figure 8). Figure 9 shows the theme drivers, with an honest split: *accuracy* and *trust* drive the negative class in **both** sources, while *automation* and *support* spike on Trustpilot but lean positive on Reddit. That divergence is itself a finding: Reddit skews toward hosts and builders who value automation, Trustpilot toward end customers who resent it. Same technology, opposite reception, depending on who meets it and whether they chose it.

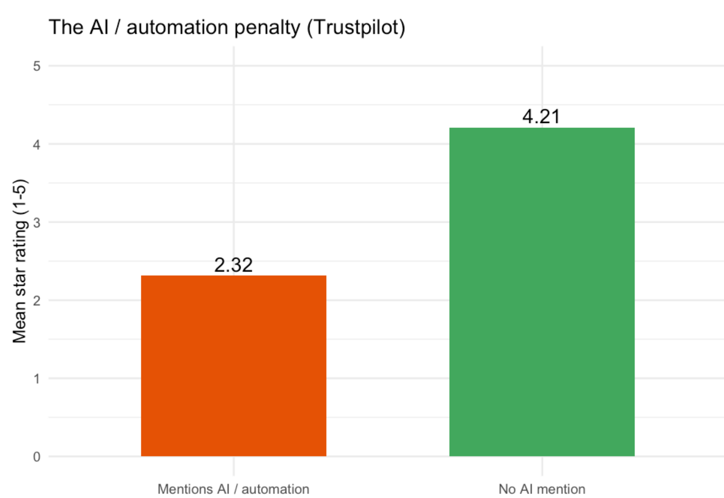

Figure 8: The AI and automation penalty on Trustpilot.

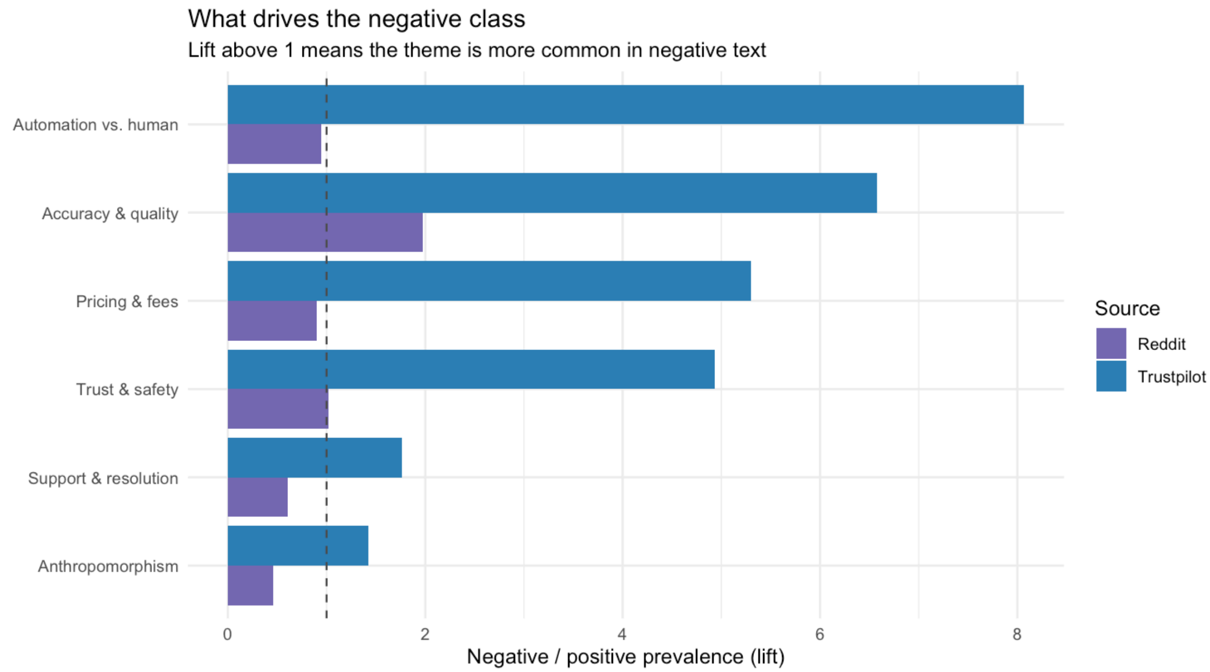

Figure 9: Themes that drive the negative class. Lift above 1 means more common in negative text.

5 Key differences between the two classes

Across the methods, the two classes separate along one axis: whether the automation **helps** the user or **blocks** them.

Positive class	Negative class
Automation that works, successful hosting	Accuracy problems and things that “don’t work”
Trust, joy, anticipation	Anger, fear, sadness, disgust
AI as a tool the user runs	AI as a barrier between the user and a person
Automation was **chosen**	Automation was **imposed**

Acceptance is high when the user runs the automation themselves and can still reach a human; it falls when automation stands in the way. This is the practical meaning of *algorithm aversion* (Dietvorst, Simmons and Massey, 2015): people accept AI for tasks they see as mechanical and resist it for tasks they see as needing human judgement.

6 Strategic recommendations

The evidence converts into a short, ordered list of design rules. They are written for a product or marketing team deciding how an AI agent should behave.

1. **Keep a human one click away.** Automation and ‘no human’ is the strongest marker of negative reviews on Trustpilot (8x). On Reddit, where many users are hosts, automation is discussed positively, so the penalty shows up most on the customer side.
2. **Disclose the AI and admit its limits.** Accuracy and ‘doesn’t work’ complaints separate the classes in both sources (about 2x on Reddit, 6.6x on Trustpilot).

3. **Optimise for first-contact resolution.** Slow or unresolved support is a strong negative marker on Trustpilot (1.8x) and a recurring theme in the Reddit support topics.
4. **Match the persona to the role.** Anthropomorphism is welcomed when AI IS the product but reads as cold in a transaction; Reddit is most positive for AI-native services, and the Trustpilot AI penalty is a rental phenomenon.
5. **Show all-in price and refund rules up front.** Pricing and fees dominate negative rental reviews (5x on Trustpilot); a rental-specific, not AI-specific, frustration.

The first rule matters most. On the customer side (Trustpilot) the strongest signal is that being trapped with a bot, unable to reach a person, destroys acceptance. On Reddit, where many users host or build with AI, the same automation is valued, so the lesson is about who meets the AI and whether they chose it. An AI agent should be **transparent** (labelled as AI, honest about limits), **escapable** (a human is one click away) and **role-matched** (personable when it is the product, efficient and unobtrusive when it is a transactional add-on).

7 Methods, tools and limitations

The analysis is written in R. Preprocessing uses tidytext and textstem; sentiment uses the AFINN and NRC lexicons (via tidytext and syuzhet); the network uses igraph and ggraph; topic models use topicmodels; embeddings use text2vec. The report is fully reproducible: `analysis.R` regenerates every figure and table that appears here (the AFINN lexicon downloads once through textdata).

A few limits are worth stating plainly:

- Trustpilot reviews skew positive, which is typical of opt-in review sites.
- Lexicon sentiment is an approximation of the class, so the inferred labels are a coarse instrument; the 92.8% agreement with stars shows it is good enough for classification, but exact magnitudes should be read as directional.
- The Trustpilot side covers rentals only, so the AI-native end of the role gradient rests on Reddit. The rental contrast, the core of the sharing economy, is the part validated across both sources.
- Trust appears mostly implicitly (bot or human, automated, algorithm), so we read it through themes and emotion rather than the literal word. English only.

8 References

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